

Md Abdur Rahman

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CURRENT DESIGNATION

Senior Research Fellow, Engineering Product Development (EPD) (Nov. 2021– Present)
Pillar Singapore University of Technology and Design (SUTD),
8 Somapah Road, 487372, Singapore

Responsibilities: Responsible for the design, simulation, nanofabrication, characterization, and experimental verification of nano-optical device fabrication for various optical applications e.g., multispectral filter, photodetectors, additive manufacturing and beyond.

Primary responsibilities to work as a technical lead of an industry funded project for fabricating and commercialization of CMOS supported multispectral colors filters on industry-ready photodiodes for the application of image sensors.

Additional responsibilities include but are not limited to fabrication of thermoelectric photodetectors, 3d printing, nano-structured LED and lasing devices, and Quantum Dots (NPL) and Nanoplatelet (NPL) enabled nano-structured optical devices.

- Gained 5 years' hands-on fabrication and characterization experience at the SUTD's Class 1000 Cleanroom environment. These responsibilities include industry standard photolithography, electron beam lithography, metallization, standard lift off processes, etching methods and material characterizations tools e.g. Scanning Electron Microscopy (SEM), Ellipsometer and Micro-spectrometer.

- Organized collaborations with researchers at Korea Institute of Science and Technology (KIST, South Korea), Institute of Materials Research and Engineering (IMRE, Singapore), University of Birmingham (UB, United Kingdom) and Tokyo University of Agriculture and Technology (TUAT, JAPAN), University of Southern Denmark (SDU, Denmark) to work as a team.

RESEARCH INTERESTS

Nanofabrication: Research focused on the fabrication of multispectral color filters by employing advanced binary and gray-scale lithography methods, fabrication of novel plasmonic and planar thermoelectric photodetector in visible to mid-IR range wavelength, 3D printed structures by two photon lithography technique, nanostructured LED and Lasing device fabrication, and QD enabled nano-structured optical devices fabrication.

Optical color coatings: Research focused on developing new structural color models for aesthetic and anti-viral/anti-bacterial applications.

Magnetic storage media application: Research focused on developing and understanding the physical mechanism of growing $L1_0$ FePt magnetic thin film by magnetron sputtering method which has potential application in high density Heat Assisted Magnetic Recording (HAMR) media.

Magnetic material: Research focused on developing novel Ferrites for use of various electronic application.

EDUCATION

Ph.D. in Materials Science and Engineering (MSE), (2012–2017)
Korea Advanced Institute of Science & Technology (KAIST), South Korea

Advisor: Professor Joong Keun Park

Thesis title: *Effect of the growth-and-coalescence of epitaxial islands on the formation of continuous epitaxial $L10$ FePt thin-film and on its perpendicular magnetic domain structure*

M.Phil. in Physics, (2008–2010)
Bangladesh University of Engineering and Technology

Advisor: Professor A. K. M. Akther Hossain

Thesis title: *Study of magnetic properties of $Li_xMg_{0.4}Ni_{0.6-2x}Fe_{2+x}O_4$ ferrites*

M. Sc. in Physics, (2005–2007)
Jagannath University, Bangladesh

B. Sc. in Physics, (2000–2005)
National University (Jagannath College), Bangladesh

PROFESSIONAL
EXPERIENCE

Postdoctoral Research Associate

Korea Institute of Science and Technology (KIST)

(July 2017 – November 2021)

Lithography free Structural coloration:

- The focal point of my research involved the exploration of novel methodologies for manufacturing lithography-free structural color coatings. This entailed the intricate modeling of single-layer (composite) and multilayer thin-film designs through both analytical and numerical simulation techniques.
- These research endeavors primarily revolved around the development of structural color coatings for various applications including aesthetic enhancements for electronic devices such as mobile phones and laptops, surface embellishment, and coloration of jewelry as well as anti-bacterial and/or anti-viral surface coatings. These meticulously crafted color models also find utility in diverse optical applications such as optical filters, security printing, and beyond.
- Acquired extensive practical expertise in the realm of thin film fabrication utilizing advanced tools such as magnetron sputtering systems, thermal evaporators, and spin coaters.

Graduate Research Assistant

Department of Materials Science and Engineering, KAIST, South Korea

(Feb 2012 – Feb 2017)

Effect of the growth-and-coalescence of epitaxial islands on the formation of continuous epitaxial L10FePt thin-film and on its perpendicular magnetic domain structure

- My research on the L1₀ FePt magnetic thin film for a perpendicular magnetic recording media which has great potential as a Heat Assisted Magnetic Recording (HAMR) media. The main theme of my research has been concerned with underlying mechanism of the deposition of the epitaxial islands of L1₀-FePt thin film on MgO (002) substrate using magnetron sputtering technique and the characterization of the growth and coalescence behavior of the epitaxial islands with HRTEM method.
- Acquired 5 years' hands-on experience in operating DC/RF magnetron sputtering, Transmission Electron Microscopy (TEM), Vibrating Sample Magnetometer (VSM) analysis and X-ray Diffraction Analysis.

Graduate Teaching Assistant (Voluntary)

Microstructural Analysis in materials (MS653), MSE, KAIST, South Korea

(Feb 2016 – Feb 2017)

- Conducting the experimental section of MS653 by Training graduate students to fabricate TEM sample (thin foil) preparation by back side ion milling method and execution of laboratory section.
- Providing hands on training to graduate students on TEM operation.

Graduate Research Assistant

Department of Physics, BUET, Bangladesh

Study of magnetic properties of $\text{Li}_x\text{Mg}_{0.4}\text{Ni}_{0.6-2x}\text{Fe}_{2+x}\text{O}_4$ ferrites

(April 2008 – November 2010)

- My research focused on the thorough investigation of structural and magnetic properties of a series of polycrystalline spinel type ferrites with the nominal chemical composition $\text{Li}_x\text{Mg}_{0.4}\text{Ni}_{0.6-2x}\text{Fe}_{2+x}\text{O}_4$. The spinel state was confirmed by X-ray diffraction analysis where all ferrite specimens exhibit ferrimagnetic state at room temperature and have a low saturation field.

Graduate Teaching Assistant

Department of Physics, BUET, Bangladesh

(April 2008 – November 2010)

- Teaching Physics laboratory classes to Engineering undergraduate students.
- Invigilation in written exams.

PATENTS

(In reverse
chronological order)

1. **Title:** High-durability coloring metal member and method of producing the same.
United States Patent no. US12292587B2
Inventors: Ji Young Byun, **Md Abdur Rahman**, So Hye Choe, Seung Yong Lee, Hyungduk Ko, Song Lim Choi, Dong Kyu Kim, Kwang Deok Choi, Inuk Baek, Yun Hee Kim.
2. **Title:** Noble metal member capable of expressing vivid colors, and method of coloring surface of noble metal member.
European Patent no. EP4115764B1
Inventors: Ji Young Byun, **Md Abdur Rahman**, Sang Hoon Kim, So Hye Cho, Seung Yong Lee, Hyungduk Ko, Sung Lim Choi, Dong Kyu Kim, Kwang-Deok Choi, Inuk BAEK, Yun Hee Kim
3. **Title:** Manufacturing Method for adjusting thickness of optical film layer, method of manufacturing optical filters with F-P cavity structures, and method of manufacturing optical sensor.
Inventors: Ruitao Zheng, Sin-Heng Lim, Wui-Pin Lee, Srinivasan Lakshman Chettiar, **Md Abdur Rahman**, Joel Kwang Wei Yang
United State Patent Application no. US20240219708A1; **Application countries:** SG/TW/CN/US (filed)
4. **Title:** Thermoelectric photodetector device
Inventors: Golnoush Zamiri, Md Abdur RAHMAN, Simon Erik Wredh, Nur Qalishah Adanan, Joel Kwang Wei Yang, Zhaogang Dong.
Patent Application no. WO2025188243A8

PUBLICATIONS

(In reverse
chronological order)

Peer-reviewed journals

(Published)

1. Hidezumi Tamehiro, Akana Ueno, Hangtian Wang, **Md Abdur Rahman**, Joel KW Yang, Wakana Kubo, "Optical Wireless Power Transmission using Thermoelectric Device with Thin-Film Absorber," IEEE Journal on Wireless Power Technologies (Early Access) April 2026.
2. **Md Abdur Rahman**[†], Darya Burak[†], Kwang - Deok Choi, Nhung Vu Cam, John You En Chan, Joel KW Yang, So - Hye Cho, Sang Hoon Kim, and Ji Young Byun. "Industrial - Grade Structural Coloration with Robust and Vivid TiON - Based Structures." Advanced Optical Materials 14, no. 4 (2026): e03135. (†: Equal Contribution)
3. Nur Qalishah Adanan, Simon Wredh, **Md Abdur Rahman**, Joel K.W. Yang, and Robert E. Simpson, "Continuously tunable optical states of Sb₂Te₃," Optical Materials Express, Vol. 15, Issue 9, pp. 2289-2298 (2025).
4. Wang Zhang[†], Hongtao Wang[†], Hao Wang, Son Tung Ha, Lei Chen, Xue Liang Li, Cheng-Feng Pan, Bochang Wu, **Md Abdur Rahman**, Yujie Ke, Qifeng Ruan, Xiaolong Yang, Thomas Christensen, Joel K.W. Yang, "Nanoscale 3D Printing of Glass for Nanophotonics," Science Advances 11, eadv0267, 2025. (†: Equal Contribution)
5. Nhung C. Vu[†], **Md Abdur Rahman**[†], Syed Akhil, Emek G. Durmusoglu, T. Thu Ha Do, Pedro, L. Hernandez-Martinez, Corentin Dabard, Deepshikha Arora, Siam Uddin, Golnoush Zamiri, Hao Wang, Son Tung Ha, N. Asger Mortensen, Hilmi Volkan Demir, Joel K.W. Yang, "Tunable Multiresonant Microcavity Exciton-Polaritons in Colloidal Quantum Wells," Nanoletters. American Chemical Society, 25(15), 6109-6116, 2025. (†: Equal Contribution) (Impact factor:10.8).
6. Golnoush Zamiri[†], Simon Wredh[†], **Md Abdur Rahman**, Nur Qalishah Adanan, Cam Nhung Vu, Wang Hong Tao, Deepshikha Arora, Haruya Sugiyama, Wakana Kubo, Zhaogang Dong, Robert E. Simpson, Joel K. W. Yang, Interband Plasmonic Nanoresonators for Enhanced Thermoelectric Photodetection, Degryter: Nanophotonics, vol. 14, no. 9, 2025, pp. 1365-1373. (†: Equal Contribution) (Impact factor: 7.5)
7. Darya Burak, Jae Hyun Han, Joon Soo Hana, In Soo Kim, **Md Abdur Rahman**, Joel K. W. Yang, Byong-Kwon Ju, and So-Hye Cho, "Passivation of Photocatalytic Activity of TiO₂ through Perhydropol-ysilazane-Assisted Formation of Ultrathin SiO₂ Shell," *Nanoscale*, 2024, 16, 21960 (Impact factor:5.8)
8. Xumin Ding, Huijie Hao, Hao Wang, Xinwei Wang, Shutao Zhang, Cheng-Feng Pan, **Md Abdur Rahman**, Tong Ling, Haoyu Li, Jiubin Tan, Joel K.W. Yang, Wenlong Lu, Jian Liu, Guangwei Hu*, "Single-Shot 3D Imaging Meta-Microscope with Double-Helix Point Spread Function" *Nanoletters*. American Chemical Society, 2024, 24, 42, 13364–13373 (Impact factor:10.8)

9. Simon Wredh, Mingjin Dai, Kenta Hamada, Nur Qalishah Adanan, Golnoush Zamiri, Md Abdur Rahman, Steve Qing Yang Wu, Nancy Lai Mun, Wenhao Zai, Zhaogang Dong, Qi Jie Wang, Wakana Kubo, Joel K.W. Yang, Robert E. Simpson, "Thermally enhanced photo-thermoelectric mid-infrared detection", Proceedings of the 18th UK Heat Transfer Conference (UKHTC2024-116), Birmingham, 9-11 September 2024. (Impact factor: N/A)
10. Simon Wredh, Mingjin Dai, Kenta Hamada, **Md Abdur Rahman**, Nur Qalishah Adanan, Golnoush Zamiri, Qing Yang Steve Wu, Wenhao Zhai, Nancy Wong, Zhaogang Dong, Wakana Kubo, Qi Jie Wang, Joel K.W. Yang, Robert E. Simpson, "Thin-Film Cavities with Direct Photo Thermoelectric Conversion for Mid-Infrared Detection", Advanced Optical Materials, Wiley, 2024, 2401450 (Impact factor: 8.0)
11. Muhammad Tegar Pambudi, Deepshikha Arora*, Xiao Liang, Basudeb Sain, Anupama, Sargur Ranganath, Matthew R. Chua, Cam Nhung Vu, Golnoush Zamiri, **Md Abdur Rahman**, Hilmi Volkan Demir, Joel K. W. Yang, Lu Ding, "Deterministic Positioning of Few Aqueous Colloidal Quantum Dots", Nanoscale, Royal Society of Chemistry. 2024,16, 18339-18347 (Impact factor: 6.7).
12. **Md Abdur Rahman**, Soroosh Daqiqeh Rezaei, Deepshikha Arora, Hao Wang, Tomohiro Mori, Ser Chern Chia, John You En Chan, Parvathi Nair Suseela Nair, Siam Uddin, Cheng-Feng Pan, Wang Zhang, Hongtao Wang, Zheng Ruitao, Lim Sin Heng, and Joel K.W. Yang "Scaling Up Multispectral Color Filters with Binary Lithography and Reflow (BLR)," Degryter: Nanophotonics vol. 13, no. 19, 2024, pp. 3671-3677. (Impact factor: 7.5)
13. **Md Abdur Rahman**, Simon Wredh, Darya Burak, Joel K.W. Yang, So-Hye Cho, Seung Yong Lee, Sang Hoon Kim, and Ji Young Byun, "Widening the Gamut of Structural Colors of Gold via Insulator-Metal Bilayer Coatings" Advanced Photonics Research, Wiley, 2024, 5, 2300324. (Impact factor: 3.7)
14. Darya Burak, **Md Abdur Rahman**, Dong Chan Seo, Seung Eun Lee, Ji Young Byun, and So-Hye Cho, "In-situ Metal Deposition on Perhydropolysilazane-derived Silica for Structural Color Surfaces with Antiviral Activity, ACS Appl. Mater. Interfaces 2023, 15, 46, 54143–541. (Impact factor: 9.5)
15. Tomohiro Mori, Hao Wang, Wang Zhang, Chern Chia Ser, Deepshikha Arora, Cheng-Feng Pan, Hao Li, Jiabin Niu, **Md Abdur Rahman**, Takeshi Mori, Hideyuki Koishi, Joel K.W. Yang, "Pick-and-Place Process for the Uniform Shrinking of 3D Printed Micro- and Nano-Architected Materials, *Nature Communications*, volume 14, Article number: 5876 (2023). (Impact factor: 16.6)
16. Parvathi Nair S., Chengfeng Pan, Hao Wang, Deepshikha Arora, Qing Yang Steve Wu, **Md Abdur Rahman**, Jinghua Teng and Joel K. W. Yang, "Multimaterial 3D Prints With A Single Resin: Selective 3D Printing Of Micron-Scale Optically Opaque and Transparent Structures via Laser Decomposition, *Light: Advanced Manufacturing* 4, Article number: 25 (2023). (Impact factor: N/A)
17. **Md Abdur Rahman**, Dong Kyu Kim, Joo-Youl Huh, Ji Young Byun, "Structural colors on Al surface via capped Cu-Si₃N₄ bilayer structure", *Micromachines* 2023, 14(2), 471. (Impact factor: 3.523)
18. **Md Abdur Rahman**, Dong Kyu Kim, Jong-Kwon Lee, and Ji Young Byun, "To realize a variety of structural color adjustments via lossy-dielectric-based Fabry–Perot cavity structure", *Degryter: Nanophotonics* 2022; 11(21): 4855– 4868. (Impact factor: 7.5).
19. **Md Abdur Rahman**, Yun Hee Kim, So-Hye Cho, Seung Yong Lee, and Ji Young Byun, "Realization of structural colors via capped Cu-based F–P cavity structure", *Optics Express*, Volume 29, Issue 18, pp. 29466- 29480 (2021). (Impact factor: 3.8)
20. Yun Hee Kim, **Md Abdur Rahman**, Jong Seung Hwang, Hyungduk Ko, Joo-Youl Huh, and Ji Young Byun, Reflection color tuning of a metal–insulator–metal cavity structure using arc plasma deposition of gold nanoparticles", *Applied Surface Science*, Volume 562, 1 October 2021, 150140. (Impact factor: 6.2)
21. **Md Abdur Rahman**, S.M.K. Vivek, S.H. Kim, and J.Y. Byun, "Polarizonic-interference colouration of stainless-steel surface with Al₂O₃-Au nanocomposite thin film coating", *Applied Surface Science*, Volume 505, 1 March 2020, 144428(1)-144428(12). (Impact factor: 6.2)
22. **Md Abdur Rahman**, and J. K. Park, "On the growth and coalescence of the epitaxial islands of L10-FePt (001) thin-film." *Journal of Applied Physics*, vol. 120, Issue 16, 165302(1)- 165302(7), 27 October 2016. (Impact factor: 3.2)
23. A.K. M. Akther Hossain, **Md Abdur Rahman**, S.F.U. Farhad, Bertrand Vilquin, Hidekazu Tanaka, "Effect of Li substitution on the magnetic properties of Li_xMg_{0.40}Ni_{0.60}-2xFe₂+xO₄ ferrite", *Physica B: Condensed Matter*, Elsevier, Volume 406, Issue 8, 1 April 2011, Pages 1506-1512. (Impact factor: 2.8)

PRESENTATIONS

(In reverse
chronological order)

1. Nhung C. Vu†, **Md Abdur Rahman**†, Syed Akhil, Emek G. Durmusoglu, T. Thu Ha Do, Pedro, L. Hernandez-Martinez, Corentin Dabard, Deepshikha Arora, Siam Uddin, Golnoush Zamiri, Hao Wang, Son Tung Ha, N. Asger Mortensen, Hilmi Volkan Demir, Joel K.W. Yang, “Tunable Multiresonant Microcavity Exciton-Polaritons in Colloidal Quantum Wells,” 12th International Conference on Materials for Advanced Technologies (ICMAT), 30 June - 4 July 2025, Singapore.(†: Equal Contribution) (Poster Presentation).
2. Simon Wredh, Mingjin Dai, Kenta Hamada, **Md Abdur Rahman**, Nur Qalishah Adanan, Golnoush Zamiri, Qing Yang Steve Wu, Wenhao Zhai, Nancy Wong, Zhaogang Dong, Wakana Kubo, Qi Jie Wang, Joel K.W. Yang, Robert E. Simpson, “Direct Photothermoelectric Conversion in Thin Film Cavities for Mid-Infrared Detection”, *Asia Light Conference 2023*, Singapore. (Poster Presentation: Best Poster Award).
3. Simon Wredh, Mingjin Dai, Kenta Hamada, Nur Qalishah Adanan, Golnoush Zamiri, **Md Abdur Rahman**, Steve Qing Yang Wu, Nancy Lai Mun, Wenhao Zai, Zhaogang Dong, Qi Jie Wang, Wakana Kubo, Joel K.W. Yang, Robert E. Simpson, “Direct Photo-Thermoelectric Conversion Based Mid-Infrared Detection”, *UKHTC-2024 (18th UK Heat Transfer Conference)*.
4. **Md Abdur Rahman**, Soroosh Daqiqeh Rezaei, Deepshikha Arora, Tomohiro Mori, Ser Chern Chia, John You En Chan, Hao Wang, Wang Zhang, Parvathi Nair S., Hongtao Wang, Zheng Ruitao, Lim Sin Heng, and Joel K.W. Yang, Single Step Lithography Process to Fabricate Micron-size Transmission Color Filters, 11th International Conference on Materials for Advanced Technologies, 26 - 30 June 2023, Singapore. (Poster Presentation: Best Poster Award).
5. Darya Burak, **Md Abdur Rahman**, Dong-chan Seo, Seung Eun Lee, Ji Young Byun, So Hye Cho, “Low- Cost Structural Color Surface Decoration with Antiviral Effect” Proceedings of the *2023 IEEE 23rd International Conference on Nanotechnology (NANO) (NANO 2023)*, July 2-5, 2023, Jeju Island, Korea.
6. Yun Hee Kim, **Md Abdur Rahman**, Inuk Baek, Joo-Youl Huh, Jong Seung Hwang, Ji Young Byun, “Au 나노 입자를 사용한 MIM (Metal- Insulator- Metal) 구조의 발색 연구 (Vivid coloration via Au nanoparticle based MIM(Metal-Insulator-Metal) structure)” Poster Presentation (online), The Korean Institute of Metals and Materials meeting, (KIM+), July 15-17, 2020.
7. **Md Abdur Rahman**, K.D. Choi, I.U. Baek, Y. H. Kim, S.L. Choi, D.K. Kim, S.Y. Lee, H.D. Ko, S.H. Cho, J.Y. Byun, “Formation of vivid colors via Cu based Metal-Insulator-Metal structure, 한국표면공학회 학술발표회 초록집 (The Korean Institute of Surface Engineering), 2020.6, 19-19(1 pages) (Oral presentation)
8. **Md Abdur Rahman**, K.D. Choi, I.U. Baek, Y. H. Kim, S.H. Kim, S.Y. Lee, S.H. Cho, J.Y. Byun, “Plasmonic nano-color formation of TiON-Au and TiON-Ag thin film on Stainless Steel Substrate” Oral Presentation, The Korean Institute of Metals and Materials meeting, (KIM+), April-2019, Changwon Exhibition and Convention Center (CECO), Changwon, Republic of Korea.
9. **Md Abdur Rahman**, S.M.K. Vivek, K.D. Choi, I.U. Baek, Y. H. Kim, S.H. Kim, S.Y. Lee, S.H. Cho, J.Y. Byun, “Plasmonic nano-color formation of Al₂O₃-Au thin film on Stainless Steel Substrate.” Oral Presentation, The Korean Institute of Metals and Materials meeting, (KIM+), October, 2018, Daejeon Convention Center (DCC), Daejeon, Republic of Korea.
10. **Md Abdur Rahman**, S.M.K. Vivek, K.D. Choi, I.U. Baek, Y. H. Kim, S.H. Kim, S.Y. Lee, S.H. Cho, J.Y. Byun, “Plasmonic Nano Coloring on Stainless Steel Substrate.” Oral Presentation, The Korean Institute of Metals and Materials meeting, (KIM+), April, 2018, International Convention Center, Jeju, Republic of Korea.
11. **Md Abdur Rahman**, and Joong Keun Park, “Growth and coalescence of epitaxial islands of L10-FePt (001) thin- film”, Oral Presentation, The Korean Institute of Metals and Materials meeting, (KIM+), October, 2016 Bexco, Busan, Republic of Korea.
12. **Md Abdur Rahman**, and Joong Keun Park, “Growth of magnetic thin-film of L10 FePt and the formation of island-network structure,” Oral Presentation, The Korean Institute of Metals and Materials meeting, (KIM+), April 2016, Gyeongju, Republic of Korea.
13. **Md Abdur Rahman**, A. K. M. Akther Hossain, M. M. A. Alam Chowdhury, M.A. Mamun, S.F.U. Farhad, “Structural and Magnetic studies of Li substituted Ni -Mg ferrites”, National Conference of Physics for Development 2011, BPS, 10-11 Feb. 2011, BUET, Dhaka. (Poster Presentation).

SKILLS

Micro- and Nano-fabrication:

- Extensive experience in various nanofabrication methods: photolithography (Mask aligner), Electron Beam Lithography (Raith e-line plus, Maskless lithography (NanYTE), two-photon lithography (Nanoscribe). Dry etching tools (DRIE) and ICP RIE.
- Thin film preparation techniques: Magnetron Sputtering, E-Beam Evaporation, Arc Plasma Deposition (APD), Atomic Layer Deposition (ALD), Chemical Vapor Deposition (CVD), Plasma Enhanced Chemical Vapor Deposition (PECVD), Spin Coating, Dip Coating.

Materials Characterization:

- Spectroscopic Ellipsometry method to measure refractive indices of thin film.
- Thin film characterization methods: alpha-step measurement, Surface profilometer, X-ray Diffraction (XRD), Auger Electron Spectroscopy (AES), X-ray Photoelectron Spectroscopy (XPS), Scanning Electron Microscopy (SEM), Transmission Electron Microscopy (TEM), Atomic Force Microscopy (AFM).
- Magnetic characterization methods: MFM, VSM.
- TEM sample preparation methods: Back side ion milling (polishing, dimpling, ion milling), Focused Ion Beam.
- Spectrophotometry by Konica Minolta Spectrophotometer and Craic Microspectrophotometer
- UV-VIS Spectrophotometer and nano-spectrophotometer.
- Vacuum furnace and box furnace handling
- Solid State and Sol-gel Reaction technique for material synthesis.

Programming Language and Simulation Software:

- Programming software (Python, Matlab), CAD software (KLayout) Optical thin film designer software (OpenFilters), Optical Simulation software (Lumerical), and image processing softwares (ImageJ, Photoshop), TEM micrograph software (Gatan digital micrograph), XRD software (MDI Jade).

RESEARCH & INDUSTRY COLLABORATIONS

- Prof. Joel K.W. Yang, EPD Pillar, Singapore University of Technology and Design, Singapore.
- Prof. Hilmi Volkan Demir, Dept. of EEE, Nanyang Technological University, Singapore.
- Dr. Zhaogang Dong, Institute of Materials Research and Engineering (IMRE), Agency for Science, Technology and Research (A*STAR), 2 Fusionopolis Way, Innovis #08-03, Singapore 138634, Republic of Singapore.
- Prof. Robert E. Simpson, University of Birmingham, UK.
- Prof. N. Asger Mortensen, University of Southern Denmark, Denmark.
- Prof. Wakana Kubo, Tokyo University of Agriculture and Technology, Faculty of Engineering, Koganei, Tokyo, Japan.
- Dr. Tomohiro Mori, Industrial Technology Center of Wakayama Prefecture, Wakayama, 6496261, Japan
- Dr. Byun Ji Young, Extreme Materials Research Center, Korea Institute of Science and Technology, South Korea.
- Dr. Sang Hoon Kim, Extreme Materials Research Center, Korea Institute of Science and Technology, South Korea.
- Dr. So Hye Cho, Materials Architecture Research Center, Korea Institute of Science and Technology, South Korea.
- Dr. Seong Young Lee, Materials Architecture Research Center, Korea Institute of Science and Technology, South Korea.
- Prof. Joong Keun Park, Materials Science and Engineering, Korea Advanced Institute of Science and Technology, South Korea.
- Prof. Bertrend Vilquin, Ecole centrale de Lyon, France.
- Prof. Hidekazu Tanaka, Osaka University, Japan.
- Prof. A. K. M. Akther Hossain, Bangladesh University of Engineering and Technology.
- Lim Sin Heng, LiteON Semiconductor PTE LTD.
- Zheng Ruitao, LiteON Semiconductor PTE LTD.

HONORS AND AWARDS

1. **Review activity in:** Science Advances, SMALL, Advanced Optical Materials, Optics Express.
2. **Best Poster Award:** 11th International Conference on Materials for Advanced Technologies, 26 - 30 June 2023, Singapore.
3. **Best Poster Award as co-author:** Asia Light Conference 2023, Singapore.
4. **Media coverage for the publications:** Nanophotonics (2 publications), ACS AMI.
5. National Science and Information & Communication Technology (NSICT), Bangladesh Fellowship.
6. Fully funded scholarship for PhD program.

REFERENCES

1. Joel K.W. Yang
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